

Meeting Summary: Snowmelt Pattern Analysis Using MODIS Data

Sep 17, 2026

Meeting details

Field	Value
Meeting ID	meeting-af7569e3-b835-4719-af31-323e604d46b4
Date	September 15, 2026, 3:35 PM
Length	About 37 minutes
Participants	Dr Qualls; student team including Tyler, Chris, Joe, and Matthew

Summary

Dr. Qualls walked the team through the three-stage method behind the snowmelt pattern tool. First, find the first day of land (FTL) for every pixel in each year's daily snow images. Second, run principal component analysis (PCA) across all years to extract the recurring pattern of melt timing. Third, use that pattern to fill cloud gaps in a single day's image.

He also outlined the tool's structure: users select a watershed, while the slowest processing runs once statewide and is stored. He will share reference materials and arrange map data and hosting. He suggested the team start building the PCA pipeline on synthetic data now.

Key points

- **Data sources:** The project uses daily NASA MODIS snow data at 500 m resolution, available from 2000. MODIS is being retired. VIIRS data starts around 2012, and Dr. Qualls wants users to reach further back than VIIRS allows.
- **Threshold:** NDSI values (0 to 100) become binary snow/no-snow using a threshold. The value 10 was only an example. The original paper appears to use a different value (unclear audio, possibly 40), so this is still open.
- **Precomputation:** FTL is the slowest step, about 20 minutes in Python. It runs once per threshold over the roughly four MODIS tiles covering Idaho, with no watershed clipping, and is stored so users never trigger it. Watershed clipping happens afterward.
- **Method:** PCA keeps the melt-timing pattern that repeats most consistently across years. Clouds and other random effects change year to year, so they end up in the discarded components.

Action items

No due dates were set in the meeting.

Owner	Task	Transcript reference	Timestamp
Dr. Qualls	Upload the original paper, PowerPoint slides, and other references to the team OneDrive (background/literature folder), and tell the team when they are ready.	"I will start uploading some stuff. I'll let you know"	27:09 to 28:57; 31:59 to 32:30
Dr. Qualls	Provide sample data for the team, drawing on data archives from the lab's later journal papers.	"get some data for you to look at"	27:09 to 28:57
Project team	Build and test the PCA pipeline on a random pixels-by-years matrix before real data arrives: standardize columns, compute components, extract PC1, reshape it into the image layout.	"arbitrarily generate uh some random matrix"	29:25 to 31:58
Dr. Qualls	Contact the campus group that can supply state and watershed maps, and confirm where the tool and data will be hosted.	"I don't know yet where all of this is going to be stored."	35:21 to 36:57

Discussion highlights

Method: daily NDSI images → threshold to snow/no snow → FTL image per year → pixels-by-years matrix → PCA (keep PC1) → melt pattern image → fill cloud gaps on one day.

The first three steps are precomputed statewide.

- **Tool design:** Users select a watershed outline from a map or list, and advanced users may customize views. Asked at the project fair whether processing could be stored once, Dr. Qualls

said "yes and no": the FTL layer is stored, and watershed steps run from it. NASA distributes data in tiles, not watersheds.

- **Satellite coverage:** A polar orbit gives daily global coverage, so the study area is imaged every day. Clouds leave gaps in those images.
- **Finding FTL (hardest step):** Each year's daily images are scanned for when each pixel first changes from snow to land, giving one dated image per year. Dr. Qualls called this step "the most art." Clouds can hide the true date, and snow that returns after the first melt-out is ignored as random noise. His former graduate student's algorithm is available to build on.
- **Building the matrix:** Each year's FTL image is flattened into one column, using the same pixel order every year. The result has a row per pixel and a column per year. The original study had 17 years (17 columns), and a watershed might have about 40,000 pixels.
- **PCA:** Each column is standardized to mean 0 and standard deviation 1. A library then computes the covariance matrix and its eigenvalues and eigenvectors. Dr. Qualls compared it to daily temperatures in Moscow and Pullman, which move together through the seasons; PCA finds the one direction that captures most of that shared movement. Here, PC1 explains about 97% of the variance, and the discarded 2 to 3% is mostly cloud influence and other noise.
- **Reading the pattern:** PC1 is reshaped back into an image showing relative melt timing. Dark (low) values melt out early; light (high) values melt out later.
- **Filling cloud gaps:** On a partly cloudy day, the visible snow line is matched to a line of equal value on the PC1 image. Pixels below that value are classed as snow-free and pixels above it as snow-covered, giving a full snow map under the clouds. Dr. Qualls hopes the team gets the tool this far.

Open questions

- ☐ Which NDSI threshold should the tool use, or should users choose it?
- ☐ Where will the tool and its precomputed data be hosted?
- ☐ How will VIIRS data be combined with the longer MODIS record?
- ☐ How does the existing FTL algorithm handle snow that returns after the first melt-out?

Source: meeting transcript (transcripts.json, 80 segments). Transcription quality is poor, so quotes and names may be approximate.